

Chapter 7

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Chapter 7: Environmental and Social Impact of STEM - "STEM World Changers" Worksheet Instructions:

Hey, STEM Heroes! Welcome to the ultimate Chapter 7 worksheet, where you'll unleash your creativity and skills to tackle environmental and social challenges. This isn't just busywork—it's your chance to brainstorm, design, and pitch ideas that could save the planet or transform your community. It's short, fun, and packed with action, taking about 30-45 minutes. Grab a pen, some paper, and your boldest ideas—let's make a difference!

Part 1: Quick Think - Climate Crisis Brainstorm (5 minutes) Task: Pick a climate change issue that affects teens (like hotter summers ruining outdoor hangs or plastic trash in your fave beach spot). In 3-4 sentences, brainstorm how you'd use ALL four STEM fields (Science, Technology, Engineering, Math) to fight it. Example: For plastic trash, Science studies ocean pollution, Tech builds a cleanup app, Engineering designs a trash-trapping drone, and Math calculates cleanup costs. **Your Issue:** _____ **Your STEM Solution:**

Part 2: Mini-Design Challenge - Pick Your Impact Project (15 minutes) Task: Choose ONE focus from Chapter 7 below and sketch a quick idea or solution. No need for perfection—just doodle on this sheet or extra paper and jot 2-3 notes on how STEM makes it work. Add a teen twist (like gamifying it or making it social media-friendly).

- **Option A: Sustainability Fix** - Design a "Green School Hack" (like a no-waste lunch system). How does STEM cut the school's carbon footprint?
- **Option B: Renewable Energy Idea** - Create a portable power gadget (like a solar phone charger for camping). How does STEM power it up?
- **Option C: Social Good App** - Develop an app for a local issue (like connecting teens to free tutoring). How does STEM solve the problem? **My Choice:** _____ **Sketch Space:** (Draw here or attach a separate sheet) **STEM Notes:**

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Part 3: Pitch Power - Eco-Ethics Elevator Pitch (10 minutes) Task: Imagine pitching your Part 2 idea to a local leader or influencer to get support. Write a 2-3 sentence "hook" to grab their attention—make it punchy! Then, list ONE ethical concern (like privacy or fairness) your idea addresses. Example Hook: "Our school's drowning in trash—my Green Hack cuts waste by 50% with a fun challenge!" **My Hook:**

My Ethical Concern: _____ (e.g., ensuring equal access for all students)

Part 4: STEM Changemaker Challenge - Fun Wrap-Up (5 minutes) Task: Turn yourself into a STEM Changemaker based on this chapter! Pick a cause (climate, energy, social good) you vibed with most. Give yourself a hero name and tagline tied to your STEM mission. Example: "Solar Slayer - Charging Up a Greener Tomorrow!" **My Hero Name:** _____ **My Tagline:** _____

Bonus Fun (Optional - 5 minutes) STEM Impact Meme: Create a quick meme caption or hashtag about using STEM for good (like “#STEMSavesThePlanet” or “Engineering Earth’s Comeback!”). Write it below or share it in class for laughs and inspo! **My Meme Idea:** _____

Reflection for the Win: What’s ONE idea or topic from Chapter 7 (like renewable energy or ethical tech) that got you hyped to take action? Why does it matter to you or your crew?